



UNIVERSITY OF
LINCOLN

Session 11: Effective Teaching and Pedagogical Content Knowledge

This Session

Learning Intentions:

1. To begin to understand Pedagogical Content Knowledge (PCK).
2. To begin to understand the importance of PCK in effective teaching.

Portfolio Component Focus: Component 1 Essay
and Component 2 Portfolio

Component 1

1000 words for 50% of the mark: Effective Teaching

- In this component you will be asked to reflect on the following question drawing on relevant literature, critically discuss:

What is effective teaching in your subject?

Component 2

One side of A1 paper for 50% of the mark: Poster

- In this component, you should discuss the knowledge development of one topic from the National Curriculum and any Exam Specifications for your subject.
- You should map out the learning outcomes for the topic out from the start of **Year 7 through to and including A-level** in UK schools.
- You should consider and discuss what **knowledge, skills and understanding** is expected of the children in schools in each year group.
- You should also reflect on the implications to practice - that being what the teacher would need to know to teach the topic (think about misconceptions, learning theories and concepts).
- Conclude by summarising the key implications to practice in teaching the topic to support effective teaching and learning.

Starter

1. Write down as much as you know about what makes effective pedagogy
2. What do you know of the term pedagogy?
3. What do you think Pedagogical Content Knowledge might mean?

Pedagogical Knowledge

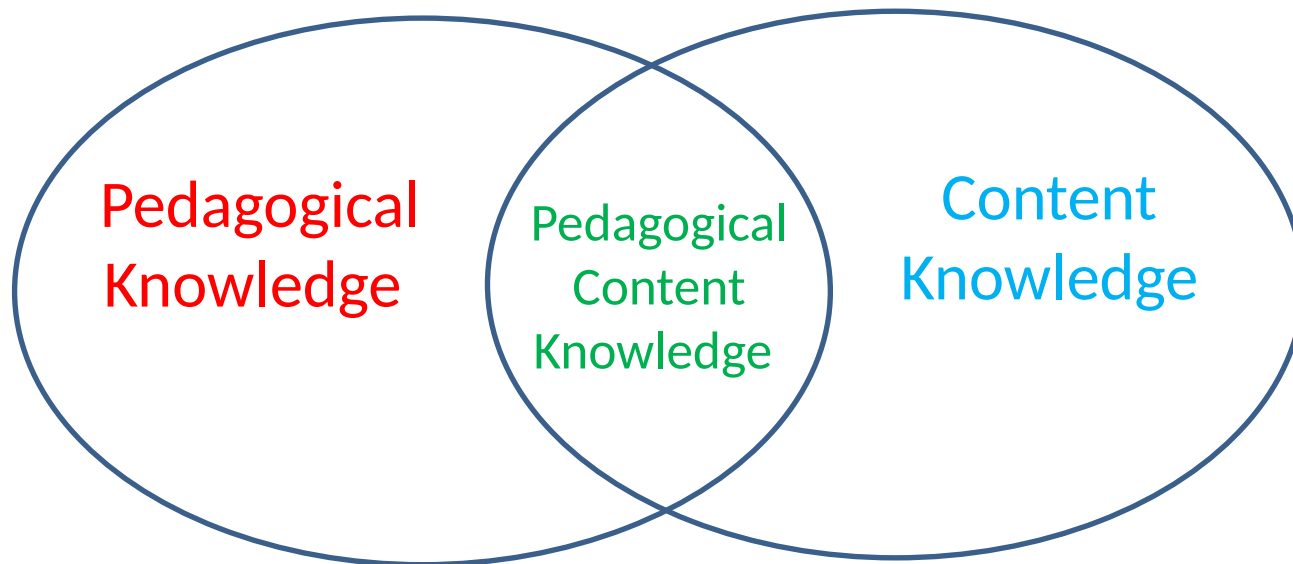
- Knowledge about teaching
- Knowledge about how students learn
- Teaching approaches
- Methods of assessment
- Knowledge of different theories about learning

Content Knowledge

- This is about subject matter
- Subject knowledge!
- No consideration about the teaching of the subject matter though.

Pedagogical Content Knowledge (PCK)

- PCK originally proposed by Lee Shulman in 1986 and 1987.
- It merges content and pedagogy into an understanding of how subject matter is taught.



PCK...

- It involves the understanding of what makes learning certain topics easy or difficult.
- That is the apparent backgrounds and prior-knowledge that students bring with them to the learning of those frequently taught lessons.
- The potential concepts and misconceptions the students hold.

PCK...

PCK is a form of practical knowledge that is used by teachers to guide their decisions and actions in subject focused classrooms. This type of practical knowledge requires, amongst other things:

1. **Knowledge of the relationship between content and the children** – a grasp of the common conceptions, misconceptions, and difficulties that children encounter when learning particular content.
2. **Knowledge of the relationship between content and the curriculum** – how to sequence, structure academic content to maximise the impact of direct teaching to children.
3. **Knowledge of the relationship between content and teaching** – the specific teaching strategies, pedagogical techniques and ways of representing, modelling and explaining knowledge that can be used to address children's learning needs in particular classroom circumstances with specific content.

Pedagogical Content Knowledge

Pedagogical Content Knowledge (PCK) is the understanding of how to put across the subject knowledge to students in a way that they would understand.

What is PCK about in your subject?

Let's put your knowledge to the test...

1. You will be teaching a 5 minute activity that you know to a non-specialist...
2. The non-specialist is then going to teach another non-specialist...
3. Who will then teach another non-specialist...
4. And so on..
5. Until at least 5 random people have been taught...
6. You then will check in with the 5th person...
7. What do you notice? Reflect in your group...

Let's Reflect

- What did you notice about the content?
- What did you notice about the participants?
- What did you notice about where it fitted into their pre-existing knowledge?

Further Reading

- Coe, R., Aloisi, C., Higgins, S., and Elliot Major, L. (2014) *What makes great teaching?* London: Sutton Trust
[Sutton Trust Document](#)
- Hanley, P., Hepworth, J., Orr, K., & Thompson, R. (2018). *Literature review of subject-specialist pedagogy*. The Gatsby Charitable Foundation. [Article Link](#)
- Husbands, C. and Pearce, J. (2012). *What makes great pedagogy? Nine claims from research*. Nottingham, NCTL.
[Article Link](#)
- [Shulman, L.S. \(1986\). Those who understand knowledge growth in teaching.](#)
[Educational Researcher, 15\(2\), 4-14.](#)
- [Stoll, L., Harris, A. and Handscomb](#)
[, G. \(2012\) Great professional development which leads to consistently great pedagogy: nine claims from research. Nottingham: NCTL](#)